```
#include<stdio.h>

void acceptMatrix( int a[10][10]  ,int m,int n)

{
    for(int i = 0; i < m; i++)
    {
        for(int j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
}

void displayMatrix(int a[10][10], int m, int n)
{
    for(int i = 0; i < m; i++)
    {
        for(int j = 0; j < n; j++)
        {
            printf("%4d", a[i][j]);
        }
        printf("\n");
    }
}

void compact(int sp[][10], int m, int n, int comp[][3])
{
int k = 1;

for(int i = 0; i < m; i++)
{
for(int j = 0; j < n; j++)
   {
     if(sp[i][j] != 0)
      {
       comp[k][0] = i;
       comp[k][1] = j;
       comp[k][2] = sp[i][j];
       k++;
            }
      }
    }

    comp[0][0] = m;
    comp[0][1] = n;
    comp[0][2] = k - 1;
```

```
}

void displaycompact(int comp[][3])
{
    int rows = comp[0][2] + 1;
    printf("\ncompact matrix:\n");
    printf("row\tcol\tvalue\n");

    for(int i = 0; i < rows; i++)
    {
        printf("%d\t%d\t%d\n", comp[i][0], comp[i][1], comp[i][2]);
    }
}
void simpletranspose(int comp[][3], int b[][3])
{
      int p =1;
      int n = comp[0][1];
    b[0][0] = comp[0][1];
    b[0][1] = comp[0][0];
    b[0][2] = comp[0][2];

 for (int k = 0; k < n; k++)
  {
  for (int j = 1; j <= comp[0][2]; j++)
        {
          if (comp[j][1] == k)
          {
          b[p][0]=comp[0][1];
          b[p][1]=comp[0][0];
          b[p][2]=comp[0][2];
          p++;
          }
       }
   }
}
void fasttranspose(int comp[][3], int b[][3])

{
    int rowTerms[20], startPos[20];
    int i, j;

    b[0][0] = a[0][1];
    b[0][1] = a[0][0];
    b[0][2] = a[0][2];

    if (a[0][2] == 0)
        return;
```

```
    for(i = 0; i < a[0][1]; i++)
        rowTerms[i] = 0;

    for(i = 1; i <= a[0][2]; i++)
        rowTerms[a[i][1]]++;

    startPos[0] = 1;

    for(i = 1; i < [0][1]; i++)
        startPos[i] = startPos[i - 1] + rowTerms[i - 1];

    for(i = 1; i <= a[0][2]; i++)
    {
        j = startPos[a[i][1]];

        b[j][0] = a[i][1];
        b[j][1] = a[i][0];
        b[j][2] = a[i][2];

        startPos[a[i][1]]++;
    }
    displayMatrix(b,k+1,3);
}

void display(int b[][3])
{
int rows = b[0][2] + 1;

      printf("\n simple transpose matrix \n:");
      printf("row\tcol\tvalue\n");

for(int i = 0; i < rows; i++)
{
      printf("%d\t%d\t%d\n", b[i][0], b[i][1], b[i][2]);
   }
}

int main()
{
      int a[20][20],a[20][20];
      int m,n,ch;
      printf("Enter Number of rows ");
      scanf("%d",&m);
      printf("Enter Number of rows ");
      scanf("%d",&n);
      acceptMatrix(a,m,n);
```

```
	displayMatrix(a,m,n);
	printf("Enter Choice");
	scanf("%d",&ch);
	switch(ch)
	{
		case 1:
			compact(a,m,n,b);
			displayMatrix(b,k+1,3);
		break;

		case 2:
			simpletranspose(a,b,n);

		break;
		case 3:
			fasttranspose(a,b);
	}

	return 0;
}
```

OUTPUT:

```
Enter rows: 2
Enter columns: 2
Enter matrix:
1 2 3 4

Original Matrix:
   1   2
   3   4

1. Simple Transpose
2. Fast Transpose
2

Fast Transpose:
Row Col Value
2   2   4
0   0   1
0   1   3
1   0   2
1   1   4
```